

Lesson 4 · Anti-differentiation, Rates and Kinematics

Learning intention

Anti-differentiate and find c , interpret rates of change in context, and move between displacement, velocity and acceleration.

Success looks like

- I can find $f(x)$ from $f'(x)$ and a point.
- I can find a rate of change and say what it means, with units.
- I can go from s to v to a , and back again.

Slide 1 · Lesson 4 · Anti-differentiation, Rates and Kinematics

Teaching line, not a question slide.

Timing: Do Now 10 min, three rounds about 20 min each, practice sheet 30 min, wrap-up 5 min.

Do Now

1 The graph of f has a maximum at $x = -1$ and a minimum at $x = 4$.

Where does the graph of f' cross the x -axis?

2 $f'(x)$ is positive for $x < 2$ and negative for $x > 2$. Does f have a maximum or a minimum at $x = 2$?

3 Differentiate $s = 5t^2 - 3t + 8$.

4 Find c if $12 = 2(3)^2 - 4 + c$.

Slide 2 · Do Now

Q1 $x = -1$ and $x = 4$

Q2 maximum

Q3 $\frac{ds}{dt} = 10t - 3$

Q4 $c = -2$

Short solution:

Q2 gradient $+$ then $-$: up then down

Q4 $12 = 18 - 4 + c$

Watch for:

Q3 the variable is t : same rule, different letter

Diagnostic:

Q1 or Q2 wrong: Lesson 3 link between turning points and zeros of f' not secure; revisit before homework

Q3 wrong: a letter other than x throws the student; say “ t works exactly like x ”

Q4 wrong: finding c will be slow today; model one more example in round 1

The one idea

Anti-differentiation **undoes** differentiation.

$$x^n \longrightarrow \frac{x^{n+1}}{n+1} + c$$

Raise the power by 1, divide by the **new** power, add $+c$.

Then use a known point to find c .

Check: differentiate your answer; you must get back $f'(x)$.

Slide 3 · The one idea

Teaching line, not a question slide.

Ask: differentiate $2x + 3$ and $2x + 10$. Both give 2, so going back we cannot know the constant: that is c .

Worked examples · anti-differentiate

1 Find $f(x)$ if $f'(x) = 6x^2 - 4x + 5$.

2 $f'(x) = 4x - 3$ and the curve passes through $(2, 7)$. Find $f(x)$.

Step 1: anti-differentiate

Step 2: substitute the point

Step 3: c

Slide 4 · Worked examples · anti-differentiate

Q1 $f(x) = 2x^3 - 2x^2 + 5x + c$

Q2 $f(x) = 2x^2 - 3x + 5$

Short solution:

Q1 $\frac{6x^3}{3} - \frac{4x^2}{2} + 5x$

Q2 $7 = 2(4) - 6 + c = 2 + c$, so $c = 5$

Watch for:

L2: $6x^2 \rightarrow \frac{6x^3}{2}$ (old power)

L1: leaving c in the final answer to Q2

Practice

1 Anti-differentiate $f'(x) = 9x^2 + 2x - 7$.

2 Anti-differentiate $f'(x) = x^3 - 6x$.

3 $f'(x) = 6x - 5$ and $f(1) = 4$. Find $f(x)$.

Slide 5 · Practice

Q1 $f(x) = 3x^3 + x^2 - 7x + c$

Q2 $f(x) = \frac{1}{4}x^4 - 3x^2 + c$

Q3 $f(x) = 3x^2 - 5x + 6$

Short solution:

Q2 $\frac{x^4}{4} - \frac{6x^2}{2}$

Q3 $f(x) = 3x^2 - 5x + c$; $3 - 5 + c = 4$

Watch for:

Q1 the constant term: $-7 \rightarrow -7x$

Q3 " $f(1) = 4$ " is the point (1, 4)

Rates of change

Write it in this order

1. Differentiate: the derivative is the rate.
 2. Substitute the time. (Given a *value*? Solve for the time first.)
 3. Units “per ...”; positive = increasing, negative = decreasing.
- 3** The water in a tank is $V = 2000 + 30t - 0.5t^2$ litres after t minutes. Find the rate of change at $t = 10$ and at $t = 40$, and say what each means.

Slide 6 · Rates of change

Q3 $t = 10$: increasing at 20 litres per minute; $t = 40$: decreasing at 10 litres per minute

Short solution:

$$\text{Q3 } \frac{dV}{dt} = 30 - t$$

Watch for:

L3: substituting into V gives the amount of water, not the rate

L4: “ -10 ” alone is not an interpretation

Worked example · find the time

4 A population is $P = 500 + 40t - t^2$ after t years. When is it growing at 10 per year?

Set the derivative equal to the rate, then solve.

Sentence frame

At $t = \dots$ the ... is increasing / decreasing at ... per ...

Slide 7 · Worked example · find the time

Q4 $t = 15$ years

Short solution:

$$\text{Q4 } \frac{dP}{dt} = 40 - 2t = 10$$

Watch for:

setting $P = 10$ instead of $\frac{dP}{dt} = 10$ (L3)

Practice

- 1 The temperature of a drink is $T = 80 - 6t + 0.1t^2$ degrees Celsius after t minutes. Find the rate of change at $t = 10$ and explain what it means.
- 2 The number of people in a queue is $N = 15t - t^2 + 20$ after t minutes. When is the number in the queue not changing?
- 3 The area of a weed patch is $A = t^2 - 4t + 10$ square metres after t weeks ($t \geq 2$). Find the rate at which the area is growing when the area is 31 square metres.

Slide 8 · Practice

Q1 -4 : cooling at 4 degrees per minute

Q2 $t = 7.5$ minutes

Q3 10 square metres per week

Short solution:

Q1 $\frac{dT}{dt} = -6 + 0.2t$, at $t = 10$: -4

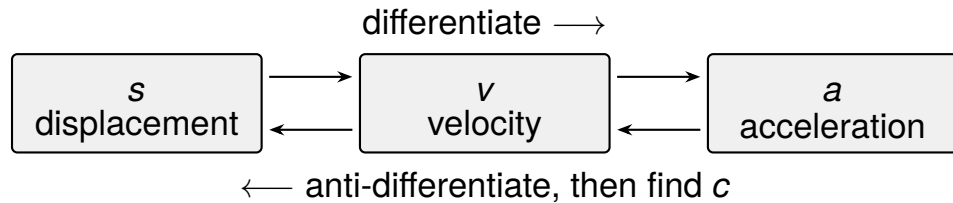
Q2 $\frac{dN}{dt} = 15 - 2t = 0$

Q3 $t^2 - 4t + 10 = 31$, so $t^2 - 4t - 21 = 0$, $t = 7$; $\frac{dA}{dt} = 2t - 4 = 10$

Watch for:

Q3 Merit: solve for the time first; reject $t = -3$ ($t \geq 2$)

Kinematics



words

maths

at rest, changes direction, maximum height

starts from the origin

starts from rest

Slide 9 · Kinematics

Row 1 $v = 0$

Row 2 $s = 0$ when $t = 0$

Row 3 $v = 0$ when $t = 0$

Short solution:

right: differentiate ($s \rightarrow v \rightarrow a$); left: anti-differentiate and find each c

Watch for:

L5: maximum height is $v = 0$, not $a = 0$

Worked examples · motion

5 A particle moves with $s = t^3 - 6t^2 + 9t$ metres after t seconds. Find its velocity at $t = 2$, the times when it is at rest, and its acceleration at $t = 3$.

6 A ball is thrown up from 1.2 m above the ground at 14 m s^{-1} . Its acceleration is -9.8 m s^{-2} . Find its maximum height.

Slide 10 · Worked examples · motion

Q5 $v(2) = -3 \text{ m s}^{-1}$; at rest at $t = 1$ and $t = 3$; $a(3) = 6 \text{ m s}^{-2}$

Q6 11.2 m

Short solution:

Q5 $v = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$; $a = 6t - 12$

Q6 $v = -9.8t + 14$; $s = -4.9t^2 + 14t + 1.2$; $v = 0$ at $t = 1.4286$

Watch for:

Q5 a negative velocity means moving backwards, not an error

Q6 two constants: $v(0) = 14$ and $s(0) = 1.2$

Practice

1 A particle moves with $s = 2t^3 - 9t^2 + 12t$ metres after t seconds. When is it at rest?

2 A cyclist starts from rest at the start line with acceleration $a = 0.6t \text{ m s}^{-2}$. Find the cyclist's velocity and distance from the start line after 5 seconds.

3 A stone is thrown straight up from the ground at 24.5 m s^{-1} ($a = -9.8 \text{ m s}^{-2}$). Find its maximum height and when it lands.

Slide 11 · Practice

Q1 $t = 1 \text{ s}$ and $t = 2 \text{ s}$

Q2 7.5 m s^{-1} ; 12.5 m

Q3 30.6 m ; lands at $t = 5 \text{ s}$

Short solution:

Q1 $v = 6t^2 - 18t + 12 = 6(t - 1)(t - 2)$

Q2 $v = 0.3t^2$ (from rest), $s = 0.1t^3$ (from the start line)

Q3 $v = 24.5 - 9.8t = 0$ at $t = 2.5$; $s = 24.5t - 4.9t^2$; $s = 0$ at $t = 5$

Watch for:

Q2 both constants are 0 here; say why

Q3 Merit: "when it lands" is $s = 0$, the second root

Ways to lose marks

L1 $+c$, then find it from the given point.

L2 Divide by the **new** power.

L3 A rate is the derivative, not the value.

L4 Say increasing or decreasing, with units.

L5 Maximum height and at rest: $v = 0$.

Slide 12 · Ways to lose marks

Teaching line, not a question slide.

Full list L1 to L6 in the handout; L6 (distance and displacement) is the homework extension.

Summary

1. $x^n \rightarrow \frac{x^{n+1}}{n+1} + c$; a point gives c .
2. Rate of change = derivative; say what the sign means, with units.
3. $s \rightarrow v \rightarrow a$: differentiate. $a \rightarrow v \rightarrow s$: anti-differentiate.
4. At rest or maximum height: $v = 0$.

Slide 13 · Summary

Teaching line, not a question slide.

Ask the student to say each line back before looking at the screen.

Practice sheet · 30 minutes

Work alone on the practice sheet.

- Questions 1 and 2: warm-up, about 6 minutes.
- Questions 3 to 8: exam questions, about 24 minutes.
- Every rate and every motion answer needs units.

Slide 14 · Practice sheet · 30 minutes

Q1 $f(x) = 4x^3 - x^2 + 3x + 4$

Q2 $s(3) = 20$ m; at rest at $t = 4$ s

Q3 (2, 23)

Q4 $f(x) = \frac{2}{3}x^3 - 3x^2 + 4x + 1$

Q5 5.7 m

Q6 -7 m s^{-1}

Q7 26.2 m above the water

Q8 (i) $P'(6) = -528$; (ii) decreasing at 528 people per hour

Short solution:

Q3 $f(x) = 2x^3 + 2.5x^2 - x + c$; $f(1) = 2.5$ gives $c = -1$

Q4 $f(3) = 12 - 27 + 18 + c = 4$, so $c = 1$

Q5 $s = 0.1t^3 + t$ with $s(0) = 0$

Q7 lands at $t = 2.6$; cliff 25.844 m; top at $t = 0.2857$

Watch for:

Q7 Excellence: needs the cliff height first; heights are above the water

Q8 (ii) people are leaving: the crowd is shrinking

Full working: practice answer version (tutor's working, not an official mark scheme)

Homework

Homework sheet: 14 questions, about 60 minutes

- Questions 1 to 11: Achieved practice.
- Questions 12 and 13: Merit.
- Question 14: Extension (Excellence).

Optional extra: StudyTime Guide, “Quick Questions” on page 29.

Slide 15 · Homework

Teaching line, not a question slide.

Answers: homework answer version (tutor’s working, not an official mark scheme).

Next lesson is the mixed practice lesson; its Do Now covers all four lessons.